YEAR 8 · SCIENCE DETECTIVES 科学侦探 · WEEK 3 OF 10 · 40 MINUTES

Who wrote the note? 纸条是谁写的?

Black ink is a mixture of dyes. Chromatography separates them into a pattern we match against reference samples, our controls.

- I can run a paper chromatogram and use reference samples to match the note ink to a pen.
- I can explain what the match shows and what it does not show.

chromatography 色谱法 · **chromatogram** 色谱图 · **mixture** 混合物 · **dye** 染料 · **solvent** 溶剂 · **dissolve** 溶解 · **reference sample** 参照样本 · **baseline** 基线 · **solvent front** 溶剂前沿 · **Rf value (Stretch)** 比移值（Rf值）

THE LESSON

Do Now	4	Fingerprint flashback: patterns 1 to 3, who was ruled out, strong or weak evidence; Today slide
Hook	3	Show the note: 'Is black ink just black?' Predict, then see a pre-test strip split into colours
Learn	7	How chromatography works; reference samples as controls; model one strip; method cloze; rules
Do	18	Set up 5 strips (5 min); Core a as they run (7 min); strips out at 12 min; match N; Stretch: Rf
Think and Reflect	8	Does the match prove who wrote it? Pair talk, frame; evidence log row 3, tracker; exit ticket

Hands-on: Groups of 4 run 5 chromatograms (pens A to D and note strip N) in 1 cm of water for about 7 min, then match N.

Collect or assess: Results and match tables (Stretch: Rf); evidence log row 3 and suspect tracker 'Ink'; exit ticket on reference samples.

SAFETY

- Water in the cups: do not drink it. Wipe up spills straight away.
- Felt-tip ink stains clothes: keep caps on when not in use.
- Wash hands after the practical.

WATCH OUT

- Water above the baseline is the most common failure: check every cup (1 cm water, baseline 2 cm up).
- 'The pen matches, so that person wrote it': two suspects carry pen B and pens can be borrowed.

Language: Frame: 'This sample matches ___ because ___.' Chinese welcome in pair talk. Traps: solvent/solution, dye/die.

Next week: Keep worksheets for Week 4 box d, spares for Weeks 7 to 8. This week, for Week 4: allergy check, powder pots.

PREP FOR ONE CLASS

before the lesson about 82 min (60 could be done by someone else); after about 23 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Buy 24 group pens (same 4 brands as pre-test); check filters for 44 strips 10 min · technician
- [ ] Print the Teacher Pre-test Record (1 copy) 2 min · teacher
- [ ] Pen pre-test: choose pen B, label A to D, fill in record, write the note 20 min · teacher

THE WEEK BEFORE

- [ ] Book printing: 24 worksheets (48 sheets) and the kit card 5 min · reprographics
- [ ] Label 24 group pens A to D with masking tape; make up 6 sets of 4 10 min · technician
- [ ] Print and laminate 6 lab cards in A5 pouches 10 min · technician
- [ ] Cut 44 filter strips (2 cm by 10 cm); make 8 N strips with pen B 10 min · technician

THE DAY BEFORE

- [ ] Optional, only if one pen set: pre-dot 24 strips A to D (optional) 10 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Set out 6 kit trays, table cards, and teacher desk items 10 min · technician

SET-UP BY STUDENTS IN THE LESSON

- [ ] Groups set up by role: baselines, dots, water to 1 cm, label cups 5 min · students

AFTER THE LESSON

- [ ] Clean-up: empty cups, dry strips flat, count pens, wipe tables, wash hands 3 min · students
- [ ] Check 24 pens, rinse cups and jugs, store strips for Week 4, reset trays 10 min · technician
- [ ] Read exit tickets; note who needs the Week 4 Do Now recap 10 min · teacher

PRINT

- Worksheet x24 (2 sheets each, double-sided)
- Chromatography lab card x6 (A5, single-sided, laminated)
- Teacher copies: kit card (A5), Pre-test Record (A4, print in Week 1)

GET OR MAKE (NEW THIS LESSON)

- Black washable felt-tips x24 (6 sets, brands A to D)
- Small clear plastic cups x30
- Kit pencils x30 (5 per group)
- Rulers, 30 cm x6
- Small jugs of water x6
- Trays x6
- Coffee-filter strips x44 (2 cm by 10 cm, cut fresh)
- Note ink strips N x8 (made with pen B)
- Paper towels x12 (2 per group)
- Laminating pouches, A5 x6
- Cloths for wiping tables (requisition: 6)
- Guillotine or scissors for cutting strips
- Paper clips, if not using tape (optional)
- Pre-dotted strips A to D x24, if one pen set (optional)

Have ready from earlier weeks: Class box with case files and logbooks, Role card and frame card sets x6, Dictionaries and writing pencils x24, Teacher pens A to D x8 (pen B), Dried pre-test strips A to D (key), Display note written with pen B, Sticky tape rolls x6 (Week 2), Masking tape roll (Week 1)